

Loops

Content notes, classwork and homework with worked C++ solutions.

Lesson 1: The for Loop

Quick recap: for loop: init, condition, update • Printing series • Running sums and products

Theory & Nuances

for loop structure

for (init; condition; update) { ... } — init runs exactly once, right before the loop starts. condition is checked before every single iteration; the moment it's false, the loop stops (its body may run zero times). update runs after each iteration's body finishes, immediately before the condition is checked again.

Printing series

Any "print a sequence" task maps directly onto choosing the right init (starting value), condition (stopping point), and update (step size, including negative steps for counting down or skipping by 2).

Running sums and products

The pattern is: declare an accumulator variable before the loop, then update it once per iteration inside the body. The starting value matters — a sum must start at 0, but a product must start at 1 (starting a product at 0 would make everything multiply out to 0).

Key Points / Common Mistakes

- Off-by-one errors are the most common loop bug — check carefully whether the boundary should use < or <=.
- The loop variable declared inside for (int i = ...) only exists inside the loop; it can't be read after it ends.
- An infinite loop happens when update never makes condition false — e.g. forgetting i++ or updating the wrong direction.

Classwork Problems

1. Print 1..N and even numbers to N

```
#include <iostream>
using namespace std;

int main() {
    int n = 10;

    cout << "Numbers 1 to " << n << ": ";
    for (int i = 1; i <= n; i++) {
        cout << i << " ";
    }
    cout << endl;

    cout << "Even numbers up to " << n << ": ";
    for (int i = 2; i <= n; i += 2) {
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

2. Sum of first N numbers

```
#include <iostream>
```

```

using namespace std;

int main() {
    int n = 20;
    int sum = 0;

    for (int i = 1; i <= n; i++) {
        sum += i;
    }

    cout << "Sum of first " << n << " numbers: " << sum << endl;
    return 0;
}

```

3. Multiplication table

```

#include <iostream>
using namespace std;

int main() {
    int n = 7;

    for (int i = 1; i <= 10; i++) {
        cout << n << " x " << i << " = " << (n * i) << endl;
    }
    return 0;
}

```

Homework Problems

1. Print N down to 1

```

#include <iostream>
using namespace std;

int main() {
    int n = 10;

    for (int i = n; i >= 1; i--) {
        cout << i << " ";
    }
    cout << endl;
    return 0;
}

```

2. Factorial with a loop

```

#include <iostream>
using namespace std;

int main() {
    int n = 6;
    long long factorial = 1;

    for (int i = 1; i <= n; i++) {
        factorial *= i;
    }

    cout << n << "! = " << factorial << endl;
    return 0;
}

```
